IDMC Milestone Assessment: MCQ Practice Set (Blueprint-Aligned)

Built from: your Assessment Preparation Blueprint and Detailed Topic-wise Bifurcation (30 MCQs = 40 marks), plus the 160-question practice bank we studied earlier.

---

1. Blueprint at a glance (MCQ section only)

Level	Topic row in blueprint	No. of Qs	Marks	Questions in this file
Easy	CDI, CAI-CDI Integration	4	4	E1 to E4
Easy	CDQ, Data Integration: Reusable Objects	4	4	E5 to E8
Easy	CAI	2	2	E9 to E10
Easy	Data Integration: Mapping	4	4	E11 to E14
Easy	Profiling, Router, Performance Tuning	4	4	E15 to E18
Easy	Data Integration: Transformations	2	2	E19 to E20
Medium	Joiner, Scheduling	1	2	M1
Medium	Agent, Agent CI Architecture, CAI-CDI Integration	2	4	M2 to M3
Medium	Taskflow	3	6	M4 to M6
Hard	Lookup, SQL Pushdown	2	4	H1 to H2
Hard	Parameterisation, Error Handling	2	4	H3 to H4
	Total	**30**	**40**	

Bonus round (B1 to B10): extra questions on bank topics that are not named in the blueprint but could still appear (Filter, Union, Sorter, Rank, CDC, Batch vs Real-time, Quarantine, Partial Pushdown, Decision branching, Snowflake).

How to use this file

1. Attempt each question first **without** looking at the answer key (Section 6).

2. Note the **Bank ref** in the key. It tells you which questions in the 160-question bank / study guide teach the same concept.
3. Rows marked **[outside bank]** (CDQ, user-defined functions, CAI process basics) were not covered in the 160-question bank. They are written from standard IDMC concepts, so check them against your course notes.
4. "Agent CI Architecture" is not defined in the bank. I treated it as Secure Agent / runtime-environment architecture (high availability, where jobs run). Confirm this against your course material.

---

2. Easy Questions (20 × 1 mark)

Row 1: CDI, CAI-CDI Integration (4)

What the exam asks: which component connects CAI to CDI, which gives runtime access to protected systems, and which decides where a job runs.

E1. A CAI process receives an API request and must then load the data into Snowflake using a CDI mapping, with no operator starting the job by hand. Which approach is correct?

- A. Keep CAI and CDI separate and let an operator start the mapping
- B. Invoke the CDI task from the CAI process using the supported CAI–CDI integration
- C. Rebuild the mapping logic inside CAI process objects
- D. Use monitoring logs to trigger the mapping

E2. A team needs reusable connectivity to an external REST API that several CAI processes will call. Which asset should be used?

- A. Mapplet
- B. Lookup transformation
- C. CAI service connector
- D. Runtime environment

E3. A CDI mapping must read an on-premises database that is not exposed to the internet. Which component normally provides the runtime connectivity?

- A. Dashboard
- B. Taskflow
- C. Mapplet
- D. Secure Agent

E4. A mapping task must run only on an approved group of Secure Agents. Which setting identifies that group?

- A. Runtime environment
- B. Project folder
- C. Connection object
- D. Mapplet

Row 2: CDQ, Data Integration: Reusable Objects (4)

What the exam asks: pick the reusable asset (mapplet, user-defined function, rule specification, dictionary) instead of copying logic.

E5. Five mappings repeat the same standardisation and validation steps, and a rule change currently needs edits in all five. What design removes the duplication?

- A. Copy the corrected transformations into every mapping
- B. Create a Secure Agent per mapping
- C. Move the shared sequence into a reusable mapplet
- D. Hardcode the logic in each mapping task

E6. *(CDQ)* A business analyst wants to define a data quality rule in a no-code way and reuse it across mappings. Which asset is designed for this?

- A. Sorter transformation
- B. Rule specification
- C. Runtime environment
- D. Command task

E7. *(CDQ)* A data quality process needs a reference list of valid values (for example valid country codes) to validate incoming data against. Which asset holds this list?

- A. Dictionary
- B. Router transformation
- C. Connection
- D. Schedule

E8. The same calculation formula is needed inside Expression transformations in many mappings. Which reusable object avoids copying the formula everywhere?

- A. Create a separate Secure Agent per mapping
- B. Copy the formula into every mapping
- C. Hardcode the formula in the mapping task
- D. Create a user-defined function and call it from the expressions

Row 3: CAI (2)

What the exam asks: what a CAI process is for, and how to speed up independent service calls.

E9. In Cloud Application Integration, which asset orchestrates steps such as calling services, assigning data and branching on conditions?

- A. Process
- B. Mapplet
- C. Mapping task
- D. Runtime environment

E10. A CAI process calls three independent external services and then runs one consolidation step. The calls currently run one after another and slow the response. What is the best change?

- A. Keep them sequential because it is simpler
- B. Cache old responses instead of calling the services
- C. Run the independent calls in parallel and join before consolidation
- D. Remove the consolidation step

Row 4: Data Integration: Mapping (4)

What the exam asks: mapping vs mapping task, parameters, connections.

E11. A mapping design is complete, but the team still needs a runnable asset with runtime settings. Which relationship is correct?

- A. Put the transformation design in the mapping task
- B. Keep the design in the mapping and use a mapping task for runnable execution settings
- C. Use the connection object to run the mapping
- D. Use the mapping for scheduling and the task for field logic

E12. Development, test and production use different file paths, but the transformation logic is identical. What avoids three hardcoded mappings?

- A. Duplicate the mapping per environment
- B. Hardcode the path in an Expression
- C. Create a new Secure Agent when the path changes
- D. Use parameters so the values can change without rewriting the mapping

E13. A new Oracle source must be added to CDI. Where are the host, service name, user and authentication details configured?

- A. A connection

- B. A taskflow decision step
- C. A mapplet
- D. A monitoring dashboard

E14. One validated mapping must be run by several jobs, each with different parameter values and schedules. What is the best arrangement?

- A. Copy the mapping for every schedule
- B. Put all transformation logic only in the task
- C. Reuse the mapping and create a separate mapping task for each execution configuration
- D. Replace the task with a connection

Row 5: Profiling, Router, Performance Tuning (4)

What the exam asks: profile before designing rules, use Router for multi-way splits, and tune using evidence and pushdown.

E15. A new customer feed has undocumented nulls and inconsistent formats. What should the developer do first?

- A. Increase Secure Agent memory
- B. Profile the source data before designing transformation rules
- C. Create a schedule
- D. Define the target load order

E16. Transactions must be separated into Domestic, International and Review paths within one mapping. Which transformation fits best?

- A. Several Aggregator transformations
- B. A Sorter transformation
- C. A Union transformation
- D. A Router transformation with separate output-group conditions

E17. A mapping moves a very large dataset into the integration engine and then filters most of it out. Which optimisation should be considered?

- A. Move even more data into the engine
- B. Disable database-side processing
- C. Push compatible filtering and processing down to the database
- D. Add a Sorter before the filter

E18. A mapping has slowed as data volume grew. What is the best first step?

- A. Review execution logs and design, then optimise pushdown, lookups and data movement
- B. Increase all compute sizes immediately
- C. Disable monitoring to reduce overhead
- D. Duplicate the mapping into smaller copies

Row 6: Data Integration: Transformations (2)

What the exam asks: choose the right transformation for a stated need.

E19. A sales mapping must calculate NET_AMOUNT from quantity, unit price and discount for every row. Which transformation is correct?

- A. Sorter
- B. Expression
- C. Router
- D. Union

E20. The target needs one row per branch containing the COUNT and AVG of many source transactions. Which transformation should be used?

- A. Lookup
- B. Router
- C. Aggregator with the branch as the group field
- D. Sorter

---

3. Medium Questions (6 × 2 marks)

Row 7: Joiner, Scheduling (1)

M1. Customer rows and order rows arrive from two different sources and must be combined into one output row when CUSTOMER_ID matches. The job must run every night at 02:00, and each environment needs its own parameter values. Which design is most appropriate?

- A. Use a Joiner transformation with the CUSTOMER_ID condition in the mapping, and attach the schedule and environment parameter values to the mapping task
- B. Use a Union transformation in the mapping and attach the schedule to the connection
- C. Use a Lookup transformation to stack both feeds and hardcode the environment values
- D. Use a Joiner transformation and create a separate Secure Agent for each environment

Row 8: Agent, Agent CI Architecture, CAI-CDI Integration (2)

M2. Two factories depend on on-premises sources for critical nightly integrations. A single Secure Agent runs every job and is a single point of failure. The business wants better availability without exposing source systems publicly, and without redesigning mappings or connections. What is the most appropriate design?

- A. Expose the databases publicly and remove the agent
- B. Create duplicate mappings tied to different hardcoded hosts
- C. Use a runtime environment containing multiple appropriately configured Secure Agents
- D. Use a taskflow decision step as a substitute for redundancy

M3. A CAI API request starts a CDI pipeline that writes business data. The pipeline can run longer than the API's request window, so the synchronous call times out. The business accepts asynchronous processing if the status can be tracked. Which decision is best?

- A. Keep the API call open until every data step finishes
- B. Trigger the CDI work asynchronously from CAI and return a trackable process status
- C. Remove CDI orchestration and start jobs manually
- D. Return success immediately and record no downstream status

Row 9: Taskflow (3)

M4. A taskflow validates incoming claim files before loading production tables. If validation fails, loading must stop and an alert path must run. If it succeeds, loading continues. Currently every step runs regardless of the result. Which fix is best?

- A. Run all steps sequentially and inspect results later
- B. Run the load and alert paths in parallel every time
- C. Use a Sorter transformation to choose the path
- D. Add a decision step that branches on the validation outcome

M5. A taskflow has six independent regional mappings followed by one consolidation task that needs every regional output. The fully sequential design misses the batch window. Which redesign keeps dependencies intact?

- A. Keep everything sequential for simplicity
- B. Run consolidation in parallel with the regional mappings
- C. Run the regional mappings in parallel, then run consolidation only after all required branches complete
- D. Remove the consolidation dependency

M6. After a data export finishes, a taskflow must run a supported system-level utility on the agent machine. The action is outside the mapping but must be controlled and tracked by the taskflow. Which step type should be used?

- A. An Aggregator transformation containing the shell command
- B. A taskflow command step
- C. A Lookup transformation as the command executor
- D. A Joiner transformation

---

4. Hard Questions (4 × 2 marks)

Row 10: Lookup, SQL Pushdown (2)

H1. A product lookup has 200,000 rows and changes once per day. The fact stream has tens of millions of rows that repeatedly request the same product keys. Uncached database lookups create excessive calls, but stale data after the daily refresh is unacceptable. Which lookup strategy is best for the daily batch?

- A. An uncached database lookup for every fact row
- B. A persistent cache that is never refreshed
- C. Replace the lookup with a Router on product keys
- D. A lookup cache refreshed for each batch, so repeated keys are resolved locally using current reference data

H2. A high-volume pipeline uses filters, joins, expressions and aggregations before loading Snowflake. Most work still runs in the integration engine, causing heavy data movement. Some expressions use functions that are not compatible with database-side execution, and business logic cannot change. Which approach is best?

- A. Force full pushdown without checking function compatibility
- B. Refactor only the incompatible expressions into pushdown-compatible logic so eligible operations run in Snowflake
- C. Disable pushdown and scale the Secure Agent
- D. Stage every transformation result in separate files before loading

Row 11: Parameterisation, Error Handling (2)

H3. A solution moves from development to test to production. Connections, schemas, file locations and credentials differ in each environment. Manual edits during promotion have caused production errors, and security-sensitive values must not be hardcoded. Which approach is most appropriate?

- A. Parameterize the environment-specific properties and supply approved runtime configuration at deployment or execution
- B. Maintain separately edited mappings for each environment
- C. Hardcode production values before promoting
- D. Store environment differences in transformation expressions

H4. A CAI payment workflow calls identity, risk, notification and ledger services. A notification outage must not roll back a successful ledger action, while an identity or risk failure must stop the payment path. Monitoring must show which branch failed and what recovery occurred. Which design is best?

- A. Apply one ignore-error rule to every service call
- B. Stop the entire workflow for any notification failure
- C. Define fault handling per service path: stop/recover for critical calls, isolated retry or alternate handling for non-critical calls
- D. Remove fault handling and rely on manual support

---

5. Bonus Round (10 extra questions)

B1. Only orders with STATUS = APPROVED should continue to the target. Which transformation is placed in the mapping path?

- A. Aggregator
- B. Joiner
- C. Filter
- D. Union

B2. North and South region feeds have the same field structure and must become one stream before loading. Which transformation is correct?

- A. Union
- B. Joiner with a fabricated key
- C. Lookup
- D. Rank

B3. A report needs the top five highest-value orders for each region, with row details preserved. Which transformation fits?

- A. Sorter alone
- B. Aggregator
- C. Router

- D. Rank configured for the top five, grouped by region

B4. A legacy consumer needs a file in customer-number order, with no grouping or filtering. Which transformation should be added before the target?

- A. Aggregator
- B. Sorter
- C. Router
- D. Lookup

B5. A source table changes continuously, but each run reloads millions of unchanged rows. Which processing approach should replace the full reload?

- A. Reload everything every cycle
- B. Change data capture / incremental processing of new and changed records
- C. Result caching
- D. A Sorter to find changes

B6. Settlement files arrive once nightly, while authorization events arrive continuously and need near-immediate handling. Which ingestion design is correct?

- A. Real-time for both
- B. Batch for both
- C. Manual load for events
- D. Batch for settlement files and real-time for authorization events

B7. A pipeline contains a few records with missing mandatory fields. Valid rows must continue and failed rows must be kept for investigation. What should the design do?

- A. Route invalid rows to a quarantine/error flow while valid rows continue
- B. Stop the pipeline on the first invalid record
- C. Discard invalid rows without any trace
- D. Load invalid rows into production

B8. Most of a mapping can be pushed to Snowflake, but one business rule has no database equivalent. Forcing it down would change the rule. What is the best compromise?

- A. Change the business rule so full pushdown works
- B. Disable all pushdown
- C. Push the compatible operations to Snowflake and keep only the incompatible rule in the integration engine
- D. Force the rule into the database anyway

B9. After validation, a taskflow must behave differently for pass, warning and critical outcomes: load, load with a warning notification, or stop and run remediation. Which is the best design?

- A. A single sequential path
- B. Decision logic with a distinct branch for each outcome
- C. Run every branch in parallel
- D. Use a Joiner transformation to choose the path

B10. Loads and dashboard queries compete for Snowflake at peak time. The queries queue heavily even though each query is fast. Which Snowflake-side decision addresses simultaneous demand?

- A. Make every IDMC mapping sequential
- B. Disable warehouse scaling
- C. Use a multi-cluster warehouse for the concurrency-heavy workload
- D. Rely only on result caching

---

6. Answer Key & Explanations

Easy

Q	Ans	Why	Bank ref
E1	**B**	CAI can invoke CDI tasks through the supported integration, giving end-to-end automation with no manual handoff.	Q97 to Q100, Q115
E2	**C**	Service connectors give reusable, centrally managed access to external APIs and services.	Q97 to Q99, Q114, Q153
E3	**D**	The Secure Agent executes jobs and reaches protected on-premises systems.	Q49 to Q52, Q117
E4	**A**	A runtime environment is the group of Secure Agent resources a task is assigned to.	Q53 to Q56, Q118
E5	**C**	A mapplet holds shared transformation logic once, so rule changes happen in one place.	Q41 to Q44, Q102
E6	**B**	Rule specifications let business users define data quality rules without code, and the rules are reusable. *[outside bank]*	none
E7	**A**	A dictionary is a reference list of valid values used by data quality assets. *[outside bank]*	none
E8	**D**	A user-defined function packages an expression once and is called from many expressions. *[outside bank]*	none (reuse theme: Q41 to Q44)
E9	**A**	A CAI process orchestrates calls, data assignment and branching. *[outside bank]*	related: Q114, Q129
E10	**C**	Independent calls run in parallel and join before the dependent consolidation step.	Q129
E11	**B**	The mapping holds the design; the mapping task is the runnable asset with runtime settings.	Q45 to Q48
E12	**D**	Parameters let paths and other values change per environment without rewriting the mapping.	Q61 to Q64
E13	**A**	A connection stores endpoint and authentication properties.	Q57 to Q60

Q	Ans	Why	Bank ref
E14	**C**	One mapping, many mapping tasks, each with its own parameters and schedule.	Q120
E15	**B**	Profile first to see nulls, patterns and distributions before designing rules.	Q1 to Q4, Q101
E16	**D**	A Router sends rows to several named output groups from one central evaluation.	Q21 to Q24, Q105
E17	**C**	Pushdown executes compatible work near the data and reduces data movement.	Q89 to Q92, Q121
E18	**A**	Tune using execution evidence, not blind scaling.	Q125, Q140
E19	**B**	Expression handles row-level calculations and derived fields.	Q5 to Q8, Q130
E20	**C**	Aggregator does grouped COUNT, AVG, SUM and similar calculations.	Q17 to Q20, Q104

Medium

Q	Ans	Why	Bank ref
M1	**A**	A Joiner combines two pipelines on a key. Schedules and per-environment parameter values belong on the mapping task, not the mapping.	Q13 to Q16, Q120
M2	**C**	Several Secure Agents inside one runtime environment give resilience with no redesign and no public exposure.	Q136
M3	**B**	Asynchronous orchestration avoids the timeout and keeps the status trackable.	Q143
M4	**D**	A decision step branches on the validation result: load on success, alert on failure.	Q73 to Q76, Q108
M5	**C**	Parallel for independent work, then gate the dependent consolidation on all required branches.	Q69 to Q72, Q109, Q133, Q144
M6	**B**	The taskflow command step handles system-level actions and reports its status.	Q81, Q84, Q126

Hard

Q	Ans	Why	Bank ref
H1	**D**	A batch-scoped cache removes repeated DB calls while the daily refresh keeps the data current.	Q134 (also Q106, Q150)
H2	**B**	Refactor only the incompatible expressions so the eligible operations can still push down. Forcing everything or disabling everything are both wrong.	Q131 (also Q141)
H3	**A**	Parameterization separates environment config from the mapping design and avoids hardcoded secrets.	Q137 (also Q107, Q145)
H4	**C**	Per-path fault handling: stop for critical services, isolate retries for non-critical ones.	Q138 (also Q110, Q159)

Bonus

Q	Ans	Why	Bank ref
B1	**C**	Filter keeps rows meeting a single condition.	Q9 to Q12
B2	**A**	Union appends compatible streams with no key matching.	Q33 to Q36, Q123
B3	**D**	Rank selects top or bottom N per group and keeps the detail. Sorter alone passes every row.	Q37 to Q40, Q124
B4	**B**	Sorter orders rows by the given keys.	Q29 to Q32, Q122
B5	**B**	CDC/incremental processing handles only changed records.	Q85 to Q88, Q111
B6	**D**	Match each workload to its latency need: batch for scheduled files, real-time for continuous events.	Q93 to Q96, Q112, Q146
B7	**A**	Quarantine rejects, keep valid rows moving, and reprocess rejects after correction.	Q128, Q142
B8	**C**	Partial pushdown keeps the exact semantics and still moves eligible work to the database.	Q160
B9	**B**	One decision step with a distinct branch per outcome.	Q149
B10	**C**	Multi-cluster warehouses address concurrent demand, which is different from making one query faster.	Q155

7. Last-minute memory hooks

Topic	Hook
CAI → CDI	CAI calls the outside world and calls CDI. No human in the middle.
Secure Agent / Runtime env	Agent = engine in your network. Runtime env = the agent group you pick.
Mapplet / UDF / Rule spec	Write once, reuse many.
Mapping vs Task	Mapping = blueprint. Task = run ticket.
Parameters	Same logic, changing values.
Profiling	Look before you cleanse.
Router	Route to many named groups.
Pushdown	Take the work to the data. If blocked, refactor only the incompatible part.
Taskflow	Sequential = depends on last. Parallel = independent, then gate. Decision = fork on result.
Lookup	Small and stable: cache. Big and daily: refresh the cache per batch. Huge and growing: reconsider pushdown.
Error handling	Per path: critical stops, non-critical retries. Bad rows go to quarantine.